

Assessing sampling designs using bayesian inference

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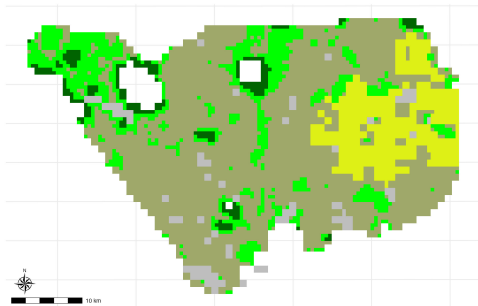
Objectives

To evaluate sampling regimes, we assessed both the inference abilities (i.e., how well the designs identify the relationship between the outcome of interest and covariates) and predictive abilities (i.e., how accurately the designs predict the outcome of interest compared to the 'true' values in space and time).

- a. Evaluate optimal sampling strategies for predicting vector distribution across a heterogenous landscape.
- b. Assess whether a single sampling design can adequately describe multiple key species.

Modeling biting rate Y_i

- $Y_i \mid \mu_i, k \sim \text{NegBinomial}(\mu_i, k)$
- $\log(\mu_{it}) = L^T \beta + s(\mathbf{x}_i) + t(\mathbf{x}_t)$
- $\beta = [\beta_{OilPalm}, \beta_{SecForest}, \beta_{PrimForest}, \beta_{OtherPlant}, \beta_{Built-up}]$

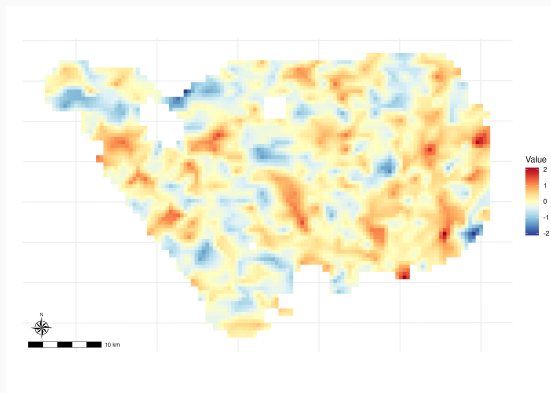


Land cover

Oil palm plantations	Other plantations
Secondary forest	Built-up
Primary forest	

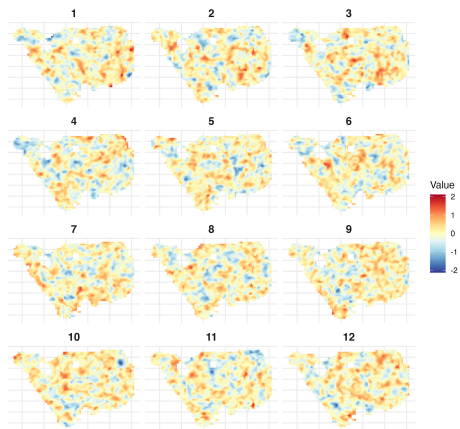
Spatial random effect

- $s(\mathbf{x}_i) \mid \theta \sim GF(0, \Sigma)$
- $\text{Cov}(s(\mathbf{x}_i), s(\mathbf{x}_j)) = \frac{\sigma^2}{2^{\nu-1}\Gamma(\nu)} (\kappa \|\mathbf{x}_i - \mathbf{x}_j\|)^\nu K_\nu(\kappa \|\mathbf{x}_i - \mathbf{x}_j\|)$



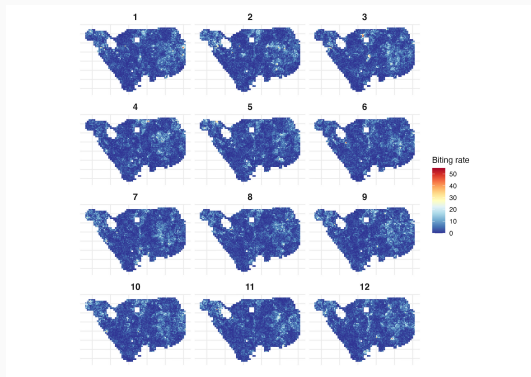
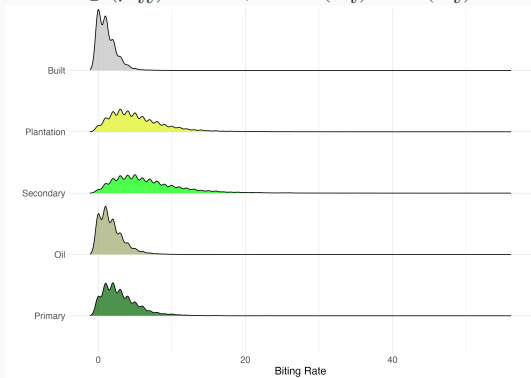
Temporal random effect

- $X_t = \varphi X_{t-1} + \varepsilon_t$



Simulated surfaces

- $Y_i \mid \mu_i, k \sim \text{NegBinomial}(\mu_i, k)$
- $\log(\mu_{it}) = L^T \beta + s(\mathbf{x}_i) + t(\mathbf{x}_t)$



Sampling strategies

Strategy	Number of sites	Strategy	Number of sites
a. fixed	5, 10, 15 fixed sites, same each month	d. stratified random	5, 10, 15 stratified random sites, equally distributed per land cover class each month
b. uniform (equally spaced)	5, 10, 15 equally spaced sites, fixed position each month	e. fixed-random	5 fixed + 5, 10 random each month
c. random	5, 10, 15 random sites, changing each month	f. fixed-stratified	5 fixed + 5, 10 stratified random